

From Markdown to TeX

A Pandoc paper pipeline — and the traps that break it

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Section 1

Agenda



Agenda

🔧 **Part 1 — The Pipeline** Why
Markdown, not raw TeX? · pandoc +
crossref + citeproc · anatomy of a source

⚙️ **Part 2 — The Build** the command,
flag by flag · one entry point: `make`

🦎 **Part 3 — War Stories** emoji &
Unicode · SVG figures · headings &
cleveref · offline citations

🎯 **Part 4 — Takeaways** a
reproducibility checklist

Section 2

The Pipeline

🤔 Why Write Markdown, Not Raw TeX?

Markdown wins ✓

- Readable plain text 📖
- Diffs are *semantic*, not macro noise 🔍
- Math stays LaTeX 📐
- One source → PDF, HTML, slides 🎯

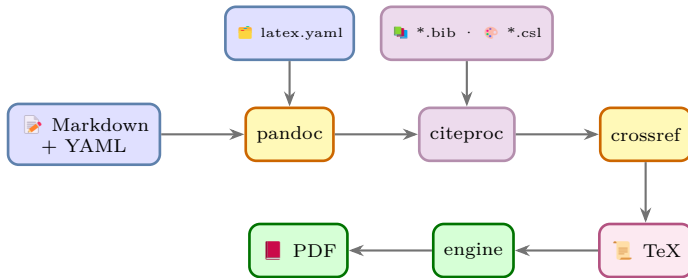
*pandoc is the bridge: Markdown **in**, TeX **out**, then a real engine compiles it.* 🏠

But TeX wins at 🏆

- Precise typesetting 📖
- Journal classes 🏛️
- Cross-refs, citations, floats 📄



The Toolchain



- **citeproc** resolves citations with a *.bib database and a *.cs1 style 
- **pandoc-crossref** resolves labelled headings, figures and equations 



Anatomy of a Pandoc-Markdown Paper

Everything is plain text — yet each @-thing is **load-bearing**:

syntax	meaning
YAML front matter	metadata: title, author, bibliography 📁
## H {#sec:foo}	a labelled heading 🏷️
@sec:foo	cross-reference → “§ 2” 🔗
[@BGT81]	citation → “[1]” 📖
math delimiters	inline & display math 📐
![alt](fig.svg)	a figure 🖼️

⚠️ *A typo in any of these becomes a ??, a missing reference, or a hard build error.*



YAML Front Matter = Document Metadata

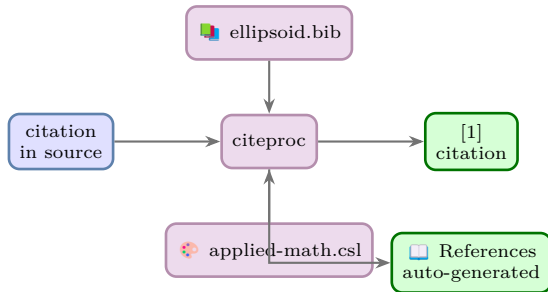
```
title: "Ellipsoid Method and the Amazing Oracles"  
bibliography: ["ellipsoid.bib", "fir-ref.bib"]  
csl: "applied-mathematics-letters.csl"  
abstract: |
```

The ellipsoid method is a powerful optimization technique ...

- title / author → \title / \author 
- bibliography → which BibTeX files to search 
- csl → the **citation style** 
- extra YAML files add **class options**: latex.yaml, crossref.yaml 
*Metadata is passed as **positional inputs** before the .md.* 



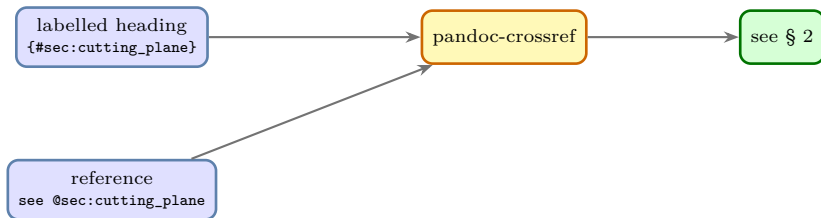
Citations: From [key] to [1]



- [key] parenthetical · key narrative · [a; b] multiple 🖋️
- Cite only keys in a **listed** .bib — otherwise the reference silently vanishes 🕒



Cross-References: From @sec:label to “§ 2”



- label prefixes: `sec` · `fig` · `eq` · `tbl` · `lst` 📌
- **the separator is file-specific** ⚠️ — `ell-review.md` uses `_`, `ell-review2.md` uses `-`
- a mismatched separator **silently breaks** the link → ?? 🕒

 Math: LaTeX Syntax, Rendered Twice

Markdown math **is** LaTeX — pandoc passes it through, and the web renders it with KaTeX:

$$\mathcal{E}(x_c, P) = \{x \mid (x - x_c)^\top P^{-1}(x - x_c) \leq 1\}$$

- in the **PDF** → real TeX, typeset by the LaTeX engine 
- in the **HTML** → KaTeX, rendered in the browser 
 - ⚠ *Keep math **inside math delimiters**. Bare Unicode (τ , x^2) in running text is what breaks `pdflatex`.*

Section 3

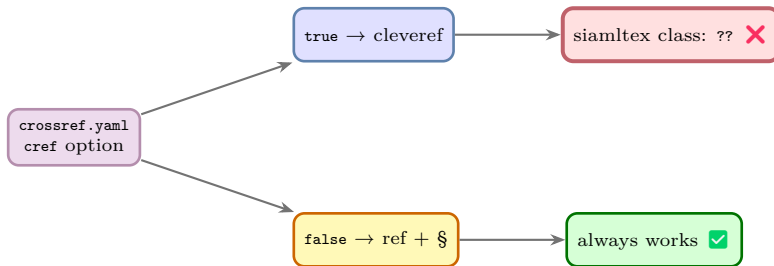
The Build



The Command, Flag by Flag

```
pandoc -F pandoc-crossref --citeproc -s -t latex -N --reference-links \  
--shift-heading-level-by=-1 \  
--csl=applied-mathematics-letters.csl \  
ell-review.yaml latex.yaml crossref.yaml ell-review.md -o ell-review.pdf
```

flag	what it does
-F pandoc-crossref	resolve heading / figure / equation labels 🏷️
--citeproc	process citations + build References 📄
-s -t latex	standalone LaTeX 📄
--shift-heading-level-by=-1	## → \section 📖
--csl=...	pick the citation style 🎨

 pandoc-crossref: cref, or plain ref?

Our policy: `cref: false` — a plain reference plus a literal § prefix. No cleveref, no ??. 🎯

Section 4

War Stories



War Story 1: Emoji & Unicode Break pdflatex

- one emoji 🏠 → ! Unicode character (U+1FA9C) not set up for use with LaTeX ❌
- but also **bare Unicode math** in running text: x^2 , $\leq$, τ , β_1 , → 🏠

broken: $x^2/4$ good: $\frac{x^2}{4} + y^2 \leq 1$

emoji & bare Unicode

pdflatex: Unicode not set up

wrap math in delimiters

builds ✓



War Story 2: \argmax Is Not a Command

- the source wrote `\argmax` — but neither LaTeX nor KaTeX defines it ❌
- result: ! Undefined control sequence — the build stops 🛑

The fix is a *real* operator ✨

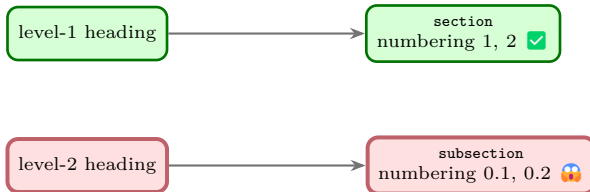
$$q_{\max} = \arg \max_{q \in \mathcal{Q}} f_0(x_0, q)$$

- `\operatorname*{arg\,max}` needs `amsmath` — pandoc already loads it ✅
- the **same bug** hid in the slide source — fix once, fix everywhere 🔁



War Story 3: Heading Levels → \section

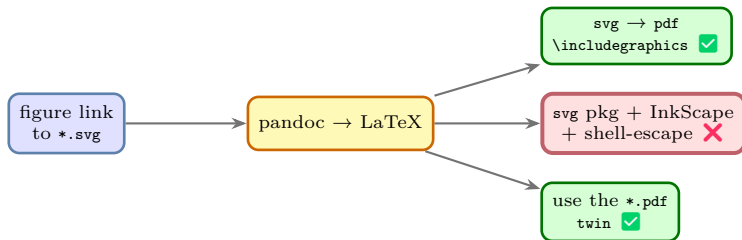
A formatter demoted # → ##; pandoc then emitted **no** \section at all.



- before: body headings at # → \section → “**1. Introduction**” ✓
- after the demotion: every heading became \subsection → “**0.1 Introduction**” 🤖
- fix **without touching the prose**: --shift-heading-level-by=-1 ✨



War Story 4: SVG Figures Need a Converter



- `pdflatex` cannot read `.svg` — pandoc wants `rsvg-convert` (librsvg) 🏢
- missing it → the LaTeX `svg` package → needs **InkScape** + `--shell-escape` ✗
- our shortcut: reference the `.pdf` **twin** already shipped beside every `.svg` 🎯

War Story 5: rsvg-convert Drops Text

Without a fontconfig, `rsvg-convert` renders **no text at all** — geometry only.

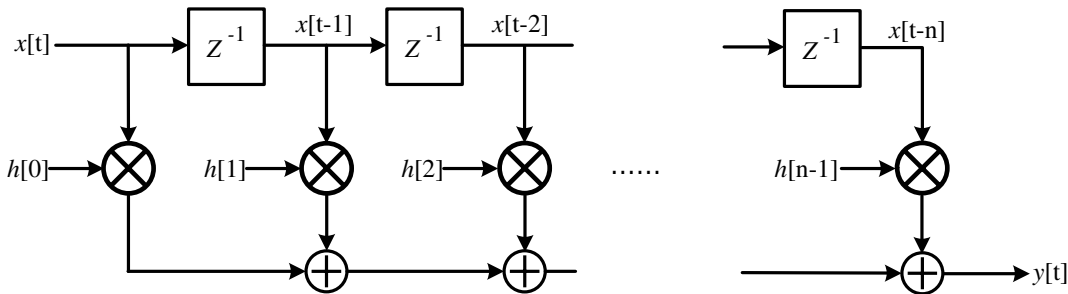
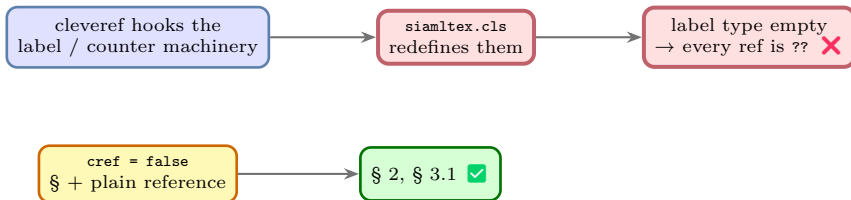


Figure 1: A typical FIR filter structure

- a converted figure can lose every label ($h[0]$, $x[t-1]$, $y[t]$) 🏷️
- a picture can look “present” yet be **informationally empty** 🕒

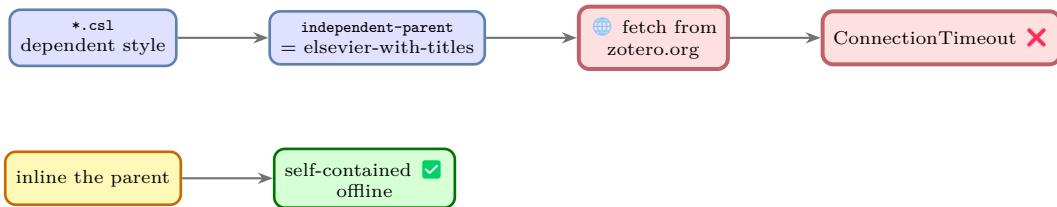
War Story 6: cleveref vs siamltex.cls



- proven with a **minimal test**: `article` + `cleveref` → fine; `siamltex` + `cleveref` → ?? 📄
- **latest pandoc** does not help — it is the **class**, not the version 📌



War Story 7: Citations That Phone Home



- that CSL file was only a **stub** — its citation / bibliography logic lived in a parent 🧩
- every fresh machine / CI fetches it → flaky and **non-reproducible** 🎲
- fix: **vendor the parent** → `--citeproc` works **offline** (verified with a dead proxy) 🔒

Section 5

Engineering the Build



One Entry Point: the Makefile

target	produces
make paper	ell-review.pdf 
make html	ell-review.html (needs katex/) 
make slides	cutting_plane.pdf, ell.pdf (xelatex) 
make clean	remove temp.* and aux files 

```
paper: ell-review.pdf
```

```
ell-review.pdf: ell-review.md
```

```
pandoc -F pandoc-crossref --citeproc -s -t latex -N \  
  --reference-links --shift-heading-level-by=-1 \  
  ell-review.yaml latex.yaml crossref.yaml ell-review.md -o $@
```

*The Makefile records the **flags and the reasons** — no more tribal knowledge.* 



Make on Windows: Two Gotchas

1. Recipes are exec'd directly 🐞

- a *simple* recipe runs **without a shell**
- `rm` is not an `.exe` here → `CreateProcess` failed ❌
- force a shell with a metacharacter ✅

clean:

```
-@rm -f temp.tex temp.pdf ; true
```

2. Line endings 📏

- LF in the repo, CRLF on disk
- keeps diffs **logical**, not whole-file 🔍

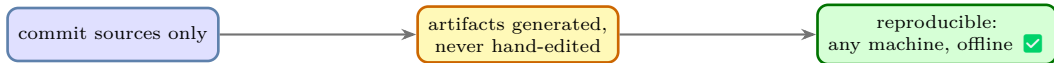
3. Version-matched tools ⚖️

- `pandoc` and `pandoc-crossref` must agree
- a mismatch warns and can misbehave silently 🙊



Repo Hygiene

- **editor backups** (*~, *.un~) — untrack and .gitignore them 🗑️
- **stale outputs** (ell-review.tex, main-diff.*) — generated, not source 📄
- **one self-contained CSL** — no network at build time 📄
- **one linter, one config** — two disagreeing linters equal *zero* linters 🗑️
- **refresh the agent guide** — write the *why* while it is fresh 🧠



Verify the Artifact, Not the Exit Code

```
make paper                                # exit 0 is necessary, not sufficient
pdfinfo ell-review.pdf                   # page count
pdftotext ell-review.pdf - | grep -c '??' # unresolved references
```

- ?? count → did the cross-references resolve? 
- extracted text → are the **figure labels** present? (catches dropped SVG text) 
- page count → did the build actually **change** anything? 
- a tiny **CI job** running `make paper` would have caught the stale PDF 



Reproducibility Checklist

Dependencies

- pandoc **and** pandoc-crossref, version-matched 
- a LaTeX engine (`pdflatex` / `xelatex`) 
- an SVG converter **or** committed PDF twins 

Sources

- a self-contained CSL (offline) 
- ASCII math, no emoji (or `xelatex`) 
- exactly one crossref policy 

same inputs

same command

same output





Key Takeaways

The pipeline 🏠

- pandoc + crossref + citeproc turns Markdown into journal-ready TeX 📖
- one **make** target beats a notebook of shell history 📁

The habit 🧠

- inspect the **artifact**, not the exit code 🔍
- vendor **everything** remote — a build must work offline 📴

The traps 🏠

- non-ASCII (emoji, bare Unicode) kills pdf_latex 💀
- old classes can break cleveref → every reference becomes ?? 🧩
- “standard-looking” macros may not exist (\argmax) 🐉



References & Further Reading

- **pandoc** — the user's guide & MANUAL: <https://pandoc.org/MANUAL.html> 
- **pandoc-crossref** — labels for sections, figures, equations 
- **CSL** — **Citation Style Language**: <https://citationstyles.org> 
- **cleveref** — and why older document classes can fight it 
- the paper: “**Ellipsoid Method and the Amazing Oracles**” 
*Assembled for the `ellipsoid-method` repo; the full checklist lives in **issue #4**.* 



Q&A

From Markdown to TeX — A Pandoc Paper Pipeline

Questions? Discussion? 



Thank You

Same input, same command, same output. 🎯

Slides built with Beamer · TikZ 🧩 · LuaLaTeX 📐 · Nord 🌙